

Sl.

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 260: Digital Logic Design

Examination: **Quiz 2 [CO2]**
Duration: 25 Minutes

Semester: Spring 2025
Full Marks: 15

Answer the following questions. You **MUST** show your workings/calculations where applicable.
Figures in the right margin indicate marks.

Name:	ID:	Section:
-------	-----	----------

1.	Let $F = C + (\bar{A} (A + \bar{B}))$ (a) Draw the circuit diagram of the given function using NAND gates only. <i>(Do not change or simplify the given expression).</i>	3
2.	Simplify the following Boolean equation using the laws of boolean algebra: $F(W, X, Y, Z) = \overline{XZY} + \overline{WXZ} (YZ + WY\bar{Z})$	7
3.	Let $F(A, B, C, D) = AB + C'D' + A'D$ (a) Derive the canonical POS from the given equation using the laws of Boolean Algebra. Write the maxterm numbers at the very last line (use Π notation). <i>(You CANNOT use truth-table or SOP to solve this).</i>	5

Sl.

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 260: Digital Logic Design

Examination: **Quiz 2 [CO2]**
Duration: 25 Minutes

Semester: Spring 2025
Full Marks: 15

Answer the following questions. You **MUST** show your workings/calculations where applicable.
Figures in the right margin indicate marks.

Name:	ID:	Section:
-------	-----	----------

1.	Let $F = C (\bar{A} + \bar{A}\bar{B})$ (a) Draw the circuit diagram of the given function using NOR gates only. <i>(Do not change or simplify the given expression).</i>	3
2.	Simplify the following Boolean equation using the laws of boolean algebra: $F(A, B, C, D) = \overline{BDC} + \overline{ABD} (BC + \overline{ACD})$	7
3.	Let $F(W, X, Y, Z) = WX + Y'Z' + W'Z$ (a) Derive the canonical POS from the given equation using the laws of Boolean Algebra. Write the maxterm numbers at the very last line (use Π notation). <i>(You CANNOT use truth-table or SOP to solve this).</i>	5